IOT DISCOVERY PROTOCOLS: A MQTT-DISCOVERY AND UPNP COMPARISON

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Available at https://github.com/DaniDF/MQTT-Discovery-vs-UPnP/blob/main/docs/MQTTDiscovery_vs_UPnP.pdf

BACKGROUND

IoT: *A global infrastructure for the information society, enabling advanced services by interconnecting (physical and virtual) things based on existing and evolving interoperable information and communication technologies.* [I.T.U.]

Large number of (different) devices require **DISCOVERY** solutions.

WHERE	**HOW**	**WHAT**
are the devices?	*to interact with them?*	*can they offer?*

MQTT-D VS UPNP

	UPnP	MQTT-D
Architecture	Decentralized (P2P)	Centralized (Client-Broker)
Transport Protocol	UDP Multicast (HTTP over UDP)	TCP/IP
Network Scope	Local (LAN, Time-to-Live = 2)	Global (not limited)
Data Format	XML	JSON

PROPOSITION

Stress **UPnP** and **MQTT-D discovery** and **communication performances** in a highly **dense network** environment.

METRICS

Latency

Time needed for the execution of the first interaction between control-point and device

Robustness

The ratio between discovered devices and total deployed units

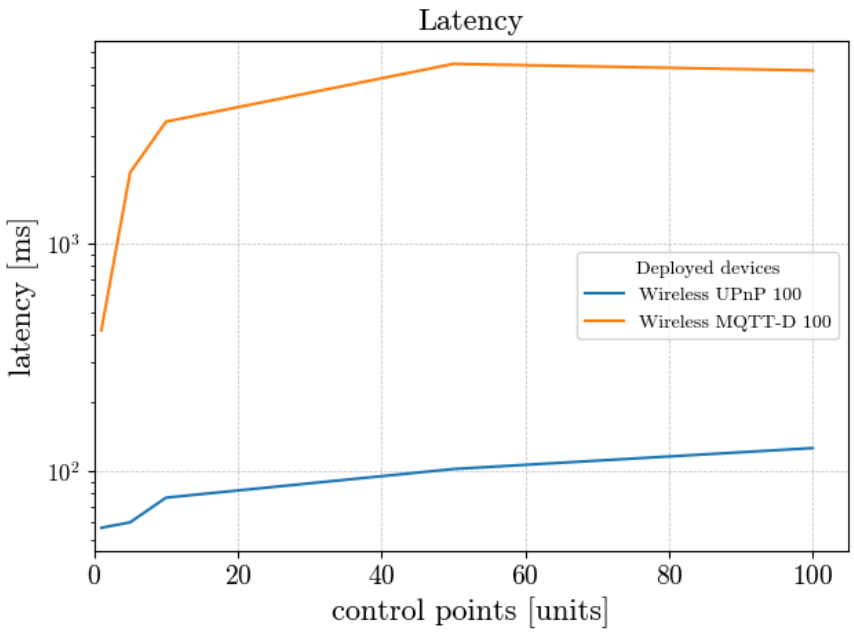
EXPERIMENT SETUP

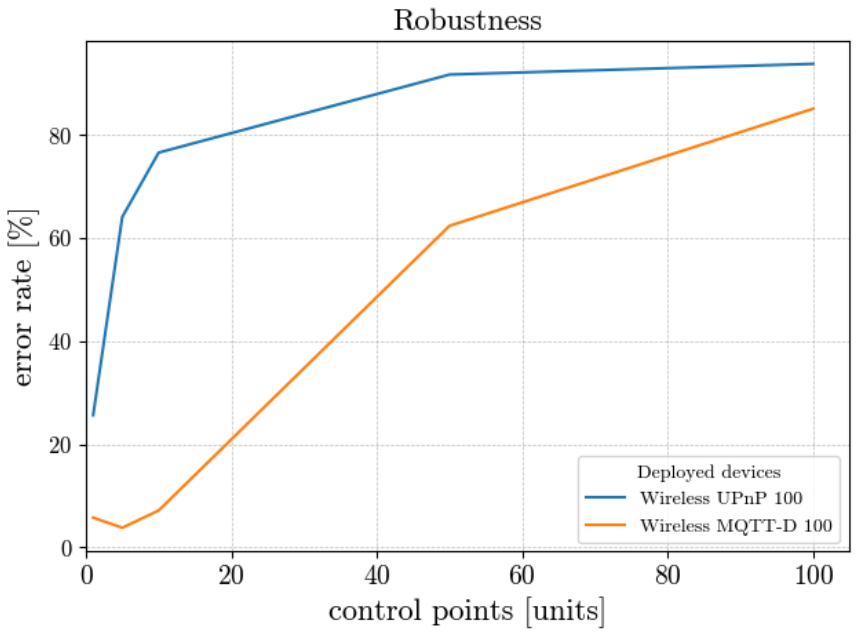
TODO: foto setup

Two nodes: one playing the role of ***device*** the other the role of ***control-point*** (a third plays the role of **broker** for MQTT-D).

TESTS: scaled deployment of *devices* and *control-points* **from 1 to 10** then from **10 to 100 with decal increment**. Each configuration evaluated **30 times**.

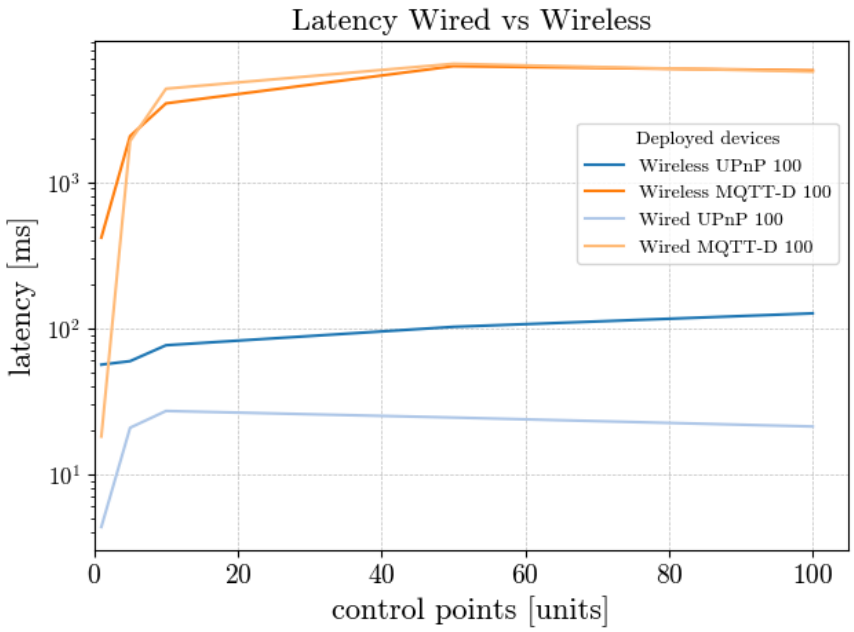
RESULTS - COMPARISON

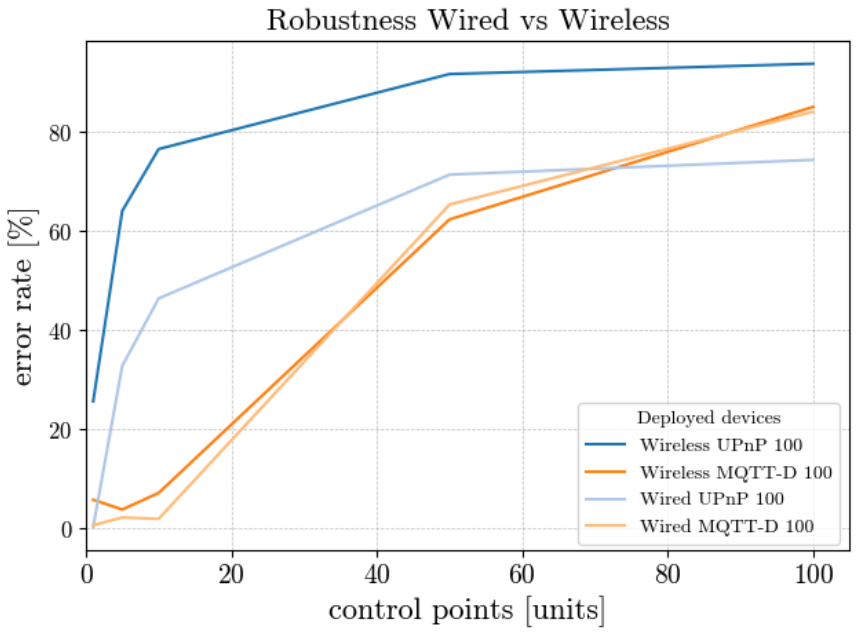




UPnP outperforms MQTT-D in terms of **latency** BUT it is **less robust**.

RESULTS - COMMUNICATION





Only **UPnP benefits** from having a **wired** connection, the performance difference in **MQTT-D** is **negligible**.

DEMONSTRATION

What we are going to see:

1. **Initial state**: Led off
2. MQTT-D:
    a. Connection to the MQTT Broker
    b. Subscribe to discovery topics
    c. Discovery of the available devices and their APIs
    d. RPC call: turn led on and measure the execution time
3. UPnP:
    a. SSDP: discovery
    b. GENA: subscribe to every device
    c. SOAP: RPC call turn led off and measure the execution time
    d. GENA: receive state update and measure the elapsed time

FUTURE WORK

Some of the still open questions:

- **Payload**: how does payload affects latency and robustness?
- **Update**: how different version of IEEE 802.11 protocols affects the measurements?
- **DTN**: how do these protocols work in disruptive networks?

SUMMARY & THANK YOU

Key Takeaways

1. UPnP better latency
2. MQTT-D better robustness
3. UPnP is more affected by the connection technology than MQTT-D

Thank you, Questions welcome